

Integrated Master of Technology in Electronics Specialization:
Embedded Systems
Batch: 2025-30

First Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL11102	Mathematics - I	4 (3-1-0)
EL11103	Basic Circuit Theory & Network Analysis	4 (3-1-0)
EL11203	Circuit Theory & Network Analysis Laboratory	1 (0-0-1)
EL11106	Digital Electronics	4 (3-1-0)
EL11206	Digital Electronics Laboratory	2 (0-0-2)
ELECTIVE COURSES-DISCIPLINE CENTRIC		
EL11105	C Programming	3 (2-1-0)
EL11205	C Programming Laboratory	2 (0-0-2)
ELECTIVE GENERIC: The students can choose following course or any other UG level generic course being run in this campus.		
EL11104	Communicative English and Technical Writing	4 (3-1-0)
Comprehensive Viva-Voce		
EL11301	Comprehensive Viva-Voce	4
VALUE ADDED (ADD-ON COURSE): It is an additional course. Its credits shall not be counted in calculating SGPA/ CGPA.		
	Students can opt for MOOC Courses as per their interest	
TOTAL CREDITS		28

Second Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL12102	Mathematics II: Set theory, Matrix theory, Integral Calculus	4 (3-1-0)
EL12105	Data Structures	4 (3-1-0)
EL12205	Data Structure Laboratory	2 (0-0-2)
ELECTIVE COURSES-DISCIPLINE CENTRIC		
EL12103	Semiconductor Device Physics & Application	4 (3-1-0)
EL12203	Semiconductor Device Laboratory	2 (0-0-2)
ELECTIVE GENERIC: The students can choose following course or any other UG level generic course being run in this campus or MOOC		
EL12106	Fundamental of Economics	4 (3-1-0)
EL12401	Student Seminar	2 (2-0-0)
Comprehensive Viva-Voce		
EL12301	Comprehensive Viva-Voce	4
VALUE ADDED (ADD-ON COURSE): It is an additional course. Its credits shall not be counted in calculating SGPA/ CGPA.		
	Students can opt for MOOC Courses as per their interest	
TOTAL CREDITS		26

Third Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL13101	Analog Electronics and Op-Amp	4 (3-1-0)
EL13201	Electronics Laboratory	2 (0-0-2)
EL13102	Java Programming	4 (3-1-0)
EL13202	Java Programming Laboratory	2 (0-0-2)
ABILITY ENHANCEMENT COURSE		
EL13105	Introduction to PCB Design	2 (2-0-0)
ELECTIVE COURSES-DISCIPLINE CENTRIC		
EL13103	Mathematics III - Probability & Statistics	4 (3-1-0)
EL13104	Electromagnetic Theory	4 (3-1-0)
EL13109	Introduction to Operating Systems	3 (3-0-0)
ELECTIVE GENERIC: The students can choose following course or any other UG level generic course being run in this campus.		
	NIL	
Comprehensive Viva-Voce		
EL13301	Comprehensive Viva-Voce	4
VALUE ADDED (ADD-ON COURSE): It is an additional course. Its credits shall not be counted in calculating SGPA/ CGPA.		
	Students can opt for MOOC	
TOTAL CREDITS		29

Fourth Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL14101	Computer Architecture & Organization	4 (3-1-0)
EL14102	Data Communication & Computer Network	4 (3-1-0)
EL14103	Signal & Systems	4 (3-1-0)
EL14203	Matlab Programming Laboratory	2 (0-0-2)
ABILITY ENHANCEMENT COURSE		
ELI4105	Python Programming	4 (3-0-0)
ELI4205	Python Programming Laboratory	2 (0-0-2)
ELECTIVE COURSES-DISCIPLINE CENTRIC		
EL14108	Mathematics IV - Real Analysis	4 (3-1-0)
ELECTIVE GENERIC: The students can choose following course or any other UG level generic course being run in this campus.		
	NIL	
Comprehensive Viva-Voce		
EL14301	Comprehensive Viva-Voce	4
VALUE ADDED (ADD-ON COURSE): It is an additional course. Its credits shall not be counted in calculating SGPA/ CGPA.		
	Students can opt for MOOC Courses as per their interest	
TOTAL CREDITS		28

Fifth Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL15101	Microprocessors & Interfacing	4 (3-1-0)
EL15201	Microprocessors Lab	2 (0-0-2)
EL15102	Analog & Digital Communication	4 (3-1-0)
EL15202	Analog & Digital Communication Lab	2 (0-0-2)
ABILITY ENHANCEMENT COURSE		
EL15191	MOOC/Other Subject	3/4 (3/4-0-0)
ELECTIVE COURSES-DISCIPLINE CENTRIC		
EL15104	Control Systems	4 (3-1-0)
EL15109	FPGA Design using Verilog	3 (3-0-0)
EL15209	FPGA Design using Verilog	1 (0-0-1)
ELECTIVE GENERIC: The students can choose following course or any other UG level generic course being run in this campus.		
Comprehensive Viva-Voce		
EL15301	Comprehensive Viva-Voce	4
VALUE ADDED (ADD-ON COURSE): It is an additional course. Its credits shall not be counted in calculating SGPA/ CGPA.		
Students can opt for MOOC Courses as per their interest		
TOTAL CREDITS		27/28

Sixth Semester

Code	Title	Credits (L T P)
CORE COURSES		
EL16101	Embedded Microcontrollers	4 (3-1-0)
EL16201	Embedded Microcontroller Lab	2 (0-0-2)
EL16103	Wireless and Mobile Communication	4 (3-1-0)
ABILITY ENHANCEMENT COURSE		
EL16104	AI and Machine Learning	4 (3-1-0)
EL16204	AI and Machine Learning Laboratory	2 (0-0-2)
ELECTIVE COURSES-DISCIPLINE CENTRIC		
EL16102	Database Management System	4 (3-1-0)
EL16202	DBMS Laboratory	2 (0-0-2)
Comprehensive Viva-Voce		
EL16301	Comprehensive Viva-Voce	4
VALUE ADDED (ADD-ON COURSE): It is an additional course. Its credits shall not be counted in calculating SGPA/ CGPA.		
Students can opt for MOOC Courses as per their interest		
TOTAL CREDITS		26

Seventh Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL17105	Wireless and Networking Protocols	4 (3-1-0)
EL17102	Digital Signal Processing	4 (3-1-0)
EL17202	Digital Signal Processing Lab	2 (0-0-2)
ELECTIVE COURSES-DISCIPLINE CENTRIC		
EL17103	Embedded Systems Design	4 (3-1-0)
EL17203	Embedded Systems Design Laboratory	2 (0-0-2)
ELECTIVE GENERIC: The students can choose following course or any other level generic course being run in this campus.		
EL17104	Linux Scripting and Networking	4 (3-1-0)
EL17204	Linux Lab	2 (0-0-2)
EL17191	MOOC	3/4 (3/4-0-0)
Comprehensive Viva-Voce		
EL17301	Comprehensive Viva-Voce	4
VALUE ADDED (ADD-ON COURSE): It is an additional course. Its credits shall not be counted in calculating SGPA/ CGPA.		
Students can opt for MOOC Courses as per their interest		
TOTAL CREDITS		29/30

Eighth Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL18101	VLSI Design Methodologies	4 (3-1-0)
EL18201	VLSI Lab	2 (0-0-2)
EL18102	Advanced Embedded Microcontroller - ARM	4 (3-1-0)
EL18202	Advanced Embedded Microcontroller Lab	2 (0-0-2)
ABILITY ENHANCEMENT COURSE		
EL18103	Real Time Systems	4 (3-1-0)
EL18203	Real Time Systems Lab	2 (0-0-2)
ELECTIVE COURSES-DISCIPLINE CENTRIC		
EL18104	IoT Fundamentals and Applications	4 (3-1-0)
EL18204	IoT Applications Laboratory	2 (0-0-2)
ELECTIVE GENERIC: The students can choose following course or any other level generic course being run in this campus.		
EL18501	Minor Project	2
Comprehensive Viva-Voce		
EL18301	Comprehensive Viva-Voce	4
VALUE ADDED (ADD-ON COURSE): It is an additional course. Its credits shall not be counted in calculating SGPA/ CGPA.		
Students can opt for MOOC Courses as per their interest		
TOTAL CREDITS		30

Ninth Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL19501	Major Project Phase - I	12

Tenth Semester:

Code	Title	Credits (L T P)
CORE COURSES		
EL19501	Major Project Phase - II	12